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Program: BSAI (Morning)

Subject: ANN & Deep Learning

Semester: 5

Lab 2: Single Layer Perceptron:

Manual Computation:

Formulas:

Sigmoid: $\sigma(x) = 1 / (1 + e^{(-x)})$

Weighted Sum: $z = w^T * x + b$

Single Neuron Computation:
LAB: 3

$$Z = (w_1x_1 + w_2x_2 + \dots + w_nx_n) + b$$
$$\{Y = f(z)\}$$

where:

- $x = \text{inputs} = [1, 2, 3]$
- $w = \text{weights} = [0.4, 0.5, 0.6]$
- $b = \text{bias} = 0.1$
- $f = \text{activation function}$
- $Y = \text{output}$

Step: 1

$$Z = (0.4 \times 1 + 0.5 \times 2 + 0.6 \times 3) + 0.1$$
$$Z = 0.4 + 1.0 + 1.8 + 0.1 \Rightarrow 3.3$$

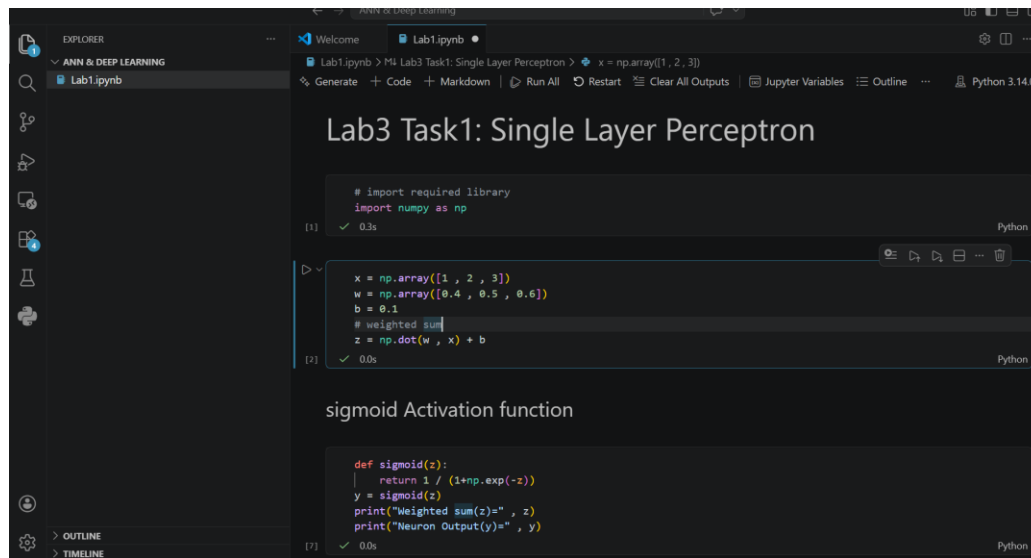
Step: 2

Applying Activation fn

Sigmoid:

$$\therefore \left\{ Y = \frac{1}{1 + e^{-z}} \right\} \Rightarrow Y = \frac{1}{1 + e^{-(3.3)}}$$
$$Y = 0.964$$

Code:



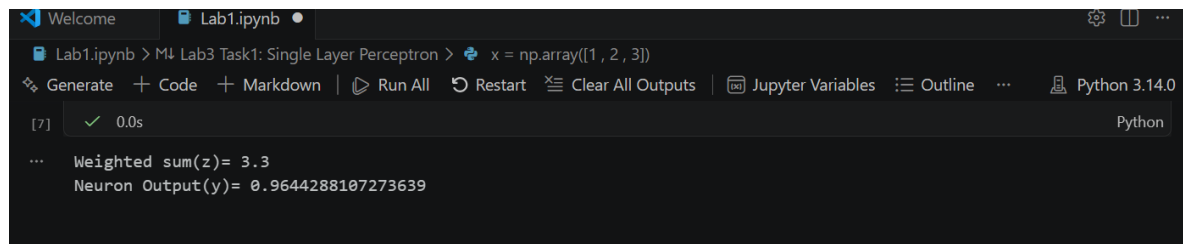
The screenshot shows a Jupyter Notebook titled 'Lab3 Task1: Single Layer Perceptron'. The code is written in Python and includes the following:

```
# import required library
import numpy as np

x = np.array([1, 2, 3])
w = np.array([0.4, 0.5, 0.6])
b = 0.1
# weighted sum
z = np.dot(w, x) + b

def sigmoid(z):
    return 1 / (1 + np.exp(-z))
y = sigmoid(z)
print("Weighted sum(z)=", z)
print("Neuron Output(y)=", y)
```

Output:



The screenshot shows the output of the code execution in the Jupyter Notebook. The output is as follows:

```
Weighted sum(z)= 3.3
Neuron Output(y)= 0.9644288107273639
```

(Manual vs Code)

Manual Output:

y=0.964

Code Output:

y=0.9644288107273639